



μProbe₅₀₀

PRECISION DEPTH-SENSING MICRO INDENTER



ACCURATE
Nanometer-scale
precision



RELIABLE
Robust design for
consistent results



ADVANCED
Depth sensing
nanotechnology



INNOVATIVE
Empowering research
and education

PRODUCT OVERVIEW

Meet μ Probe⁵⁰⁰

Precision Depth-Sensing Micro Indenter for Advanced Material Characterization

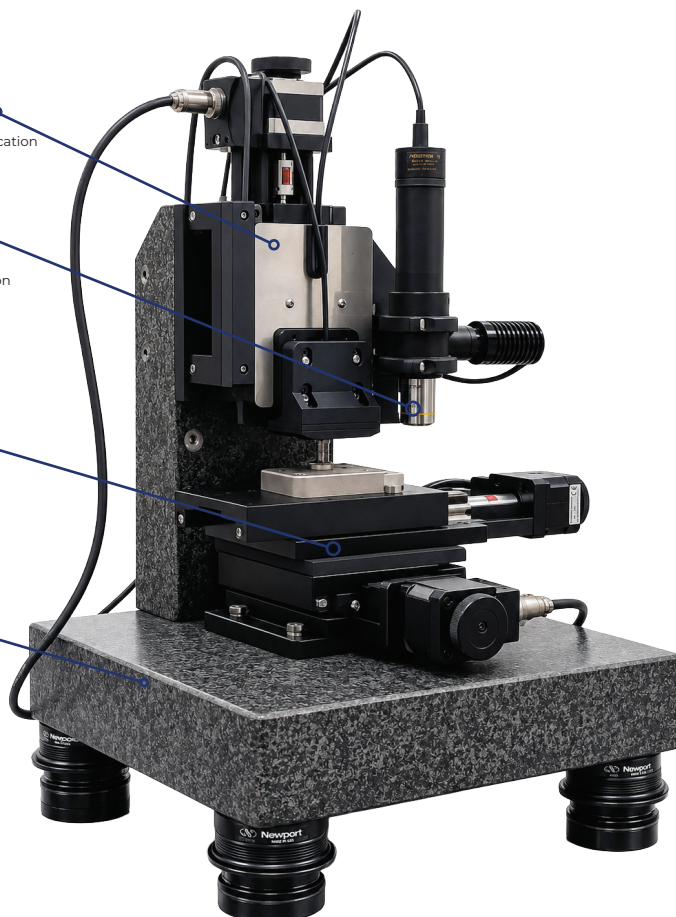
The μ Probe 500 is a research-grade depth-sensing micro indenter designed for accurate measurement of hardness, elastic modulus, and other mechanical properties across a wide range of engineering materials.

**HIGH PRECISION
ACTUATOR**
High-precision force application
and displacement sensing

**DIGITAL
MICROSCOPE**
High-resolution observation
and measurement

**X-Y-Z
STAGE**
High-accuracy positioning
and repeatability

**GRANITE
BASE**
High Stiffness &
Low vibration natural
granite for maximum
stability



500 mN

Maximum
Load Capacity



18 μ m

Maximum
Displacement



24-bit

High-Resolution
ADC



Automated

Testing &
Mapping



XYZ Stage

Motorized
Positioning



10x-40x

Optical
System

Applications

The μ Probe 500 is designed to meet the diverse needs of modern research and advanced material characterization.



Materials
Research



Thin Films &
Coatings



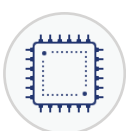
Metals &
Alloys



Polymers &
Composites



Biomaterials &
Medical Devices



Semiconductors &
Microelectronics



Advanced
Coatings



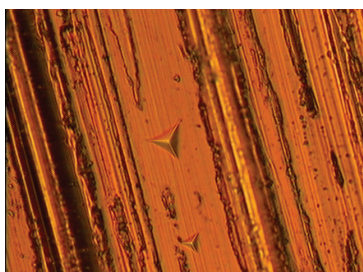
Education &
Training

Measurement Capabilities

Advanced testing modes for accurate evaluation of mechanical properties at the micro and nano scale.

01

Micro Indentation



Depth-sensing indentation to measure hardness, elastic modulus, and other mechanical properties with high accuracy.

02

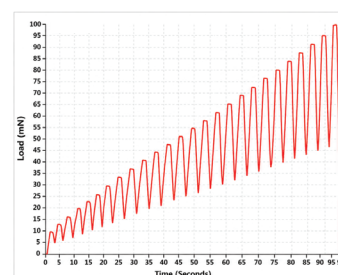
Method Automation



Automated grid indentation with stage control for repeatable and reliable results.

03

Partial Unload Testing



Instrumented partial unload technique for accurate determination of elastic modulus and reduced indentation effects.

TECHNICAL SPECIFICATIONS

SPECIFICATION	DETAILS
 Indentation Load Range	0 to 500 mN
 Maximum Displacement	18 μm
 ADC Resolution	24-bit
 Frame Stiffness	8×10^7 N/m
 Digital Control & Data Acquisition	600 MHz Embedded Processor @ 30 KHz
 Motorized Translation Stages	X-axis: 100 mm Y-axis: 50 mm Z axis: 50 mm (with 1 nm encoder resolution)
 Optics	10X to 40X



ENGINEERED FOR ACCURACY.
DESIGNED FOR DISCOVERY.



PRECISE

Nanometer-scale
measurement
accuracy



RELIABLE

Consistent
performance you
can depend on



AUTOMATED

Intelligent automation
for high-throughput
testing



INNOVATIVE

Advanced technology
driving research
forward



ADVANCED

Powerful software
for in-depth analysis
and reporting

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